Lecture plan · for the lecturer

# Adversarial examples

AI security · University · Lecture 1

**Time**
60 min

**Materials**
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

## Learning goals

- Explain why adversarial examples exist.
- Compute and run FGSM.
- Evaluate defences honestly.

## Activity: Attack, defend, attack again

1. Warm-up (5 min): Show the stop sign with stickers and ask what the classifier reads.
2. Attack (15 min): Pairs compute $\varepsilon \|w\|_1$ for a paper classifier and flip its prediction.
3. Defend (15 min): Each pair proposes a defence and states its threat model.
4. Break (20 min): Swap and design an attack that knows the other pair's defence.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

## Discussion

- Which defences survived?
- What did the threat model leave out?
- How would you report robustness?

## Exit ticket

- Name one claim you will no longer accept without an adaptive attack.

**Try it in your next lecture**
Give pairs a three-feature linear classifier on paper; one attacks it with $\varepsilon = 0.1$, the other proposes a defence, then they swap and attack the defence.

Seminar sheet · for students

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Name _____ Group _____ Date _____

**1** **A defence stops FGSM. What test comes next?**

◯ Run FGSM again with a new seed ◯ An adaptive attack designed for the defence

◯ Report clean accuracy only

**2** **Bound the change.**

A linear model has w = (1, -2, 4). $\Sigma|w_i|$ = _____, so with ε = 0.05 FGSM can change the score by _____.

**Word bank:** 0.15 · 0.35 · 7 · 3

**3** **Which belong in a sound robustness evaluation? Choose all that apply.**

☐ Iterative attacks such as PGD, not only FGSM ☐ A stated threat model and ε

☐ Reporting only clean accuracy ☐ An adaptive attack designed for the defence

**4** **A defence that stops FGSM may still fall to an adaptive attack within the same ε.**

◯ True ◯ False

**5** **Match each idea to its source.**

| | | |
|---|---|---|
| Linearity explains adversarial examples | • • • • | Madry et al., 2018 |
| Adversarial training as min-max | • • • • | Goodfellow et al., 2015 |
| Obfuscated gradients break defences | • • • • | Eykholt et al., 2018 |
| Physical stickers on stop signs | • • • • | Athalye et al., 2018 |

Answers · for the lecturer

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1 **An adaptive attack designed for the defence**

Only an attack that knows the defence tests it fairly.

2 **A linear model has w = (1, −2, 4). $\Sigma|w_i| = 7$, so with ε = 0.05 FGSM can change the score by 0.35.**

Absolute values add: 1 + 2 + 4 = 7, and 0.05 × 7 = 0.35.

3 **A stated threat model and ε; An adaptive attack designed for the defence; Iterative attacks such as PGD, not only FGSM**

Clean accuracy says nothing about robustness.

4 **True**

Weak attacks give false confidence; many such defences fell to adaptive attacks.

5 **Linearity explains adversarial examples — Goodfellow et al., 2015**
**Adversarial training as min-max — Madry et al., 2018**
**Obfuscated gradients break defences — Athalye et al., 2018**
**Physical stickers on stop signs — Eykholt et al., 2018**

Each paper is a standard reading for the topic.

✎ **For you to think about**

**When your students claim a model is secure, what evidence do you currently accept?**